Shesadree Priyadarshani

ML Research | Deep Learning, Generalization & Robustness, LLM Safety & Evaluation
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Research Interests: continual learning, generalization under distribution shift, adaptation without fine-tuning, robustness of alignment

EDUCATION

University of Florida **Aug 2025 - May 2027**
M.S. Applied Data Science, GPA: 4.0/4.0 *Gainesville, FL*
Coursework: Advanced Machine Learning, Deep Learning, Computer Vision, Optimization, Statistical Inference

National Institute of Technology Rourkela **Aug 2019 - May 2023**
B.Tech. Electrical Engineering, CGPA: 8.26/10 *Rourkela, India*

PUBLICATIONS & MANUSCRIPTS

1. J. Sena, **S. Priyadarshani**, M. Contreras, B. Gandhi, S. Siegel, S. Nerella, P. Rashidi. "A Knowledge-Injection Framework for Zero-Shot Adaptation of LLMs to Delirium Prediction." *arXiv preprint* arXiv:2607.20453, 2026.

RESEARCH EXPERIENCE

UF Intelligent Clinical Care Center (IC3), University of Florida | Gainesville, FL **Nov 2025 - Present**
AI Research Assistant - I-HEAL Research Group
- Co-developed an inference-time **knowledge-injection framework** adapting **LLaMA 3.1 8B and LLaMA 3.3 70B** to **zero-shot** ICU delirium prediction without fine-tuning or retrieval; on **3,160 MIMIC-IV admissions**, raised AUROC by **8.57 points** (8B) and cut the gap to a GPT-5.2 reference from **15.66 to 7.09** AUROC points; **random-report controls** and **SHAP attribution** showed gains came from injected knowledge, not prompt length.
- Built reproducible **zero-shot and few-shot evaluation** pipelines with AUROC, **calibration**, and **bootstrap CIs**.
- Developed an interpretable **multimodal deep learning** model for kidney stone classification from endoscopic images, clinical data, and CT Hounsfield units, with Shapley attribution, occlusion analysis, gradient maps, and patient-level validation.
- Combined **RAG** and **QLoRA fine-tuning**, improving grounding by **30%** and reducing hallucinations by **35%**.

Multilingual LLM Safety Study, University of Florida | Gainesville, FL **Fall 2025**
- Tested cross-lingual generalization of English-centric **safety alignment**: translated **120 jailbreak prompts** into Hindi, Tamil, Marathi, Odia (**NLLB-600M**), added native-verified cultural variants, and tested **LLaMA 3.1 70B, Mistral 7B, GPT-OSS 20B**.
- Showed alignment transfers unevenly under **language shift**: attack success rate ranged from **0.769 (Hindi)** to **0.081 (Marathi)**; Mistral 7B was most vulnerable (ASR 0.437) and failed on Odia entirely (100% error rate), while LLaMA 3.1 70B had near-zero errors.
- Built a reproducible pipeline with severity-labeled prompts, native-speaker labeling, and per-language and per-model **attack success rate**; **culturally adapted prompts** bypassed safeguards more often than literal translations.

SELECTED RESEARCH PROJECTS

Constrained Multi-Agent RL under Non-Stationary Demand | PyTorch, Ray RLlib, SUMO **Jun 2026**
- Built a simulation-based **multi-agent reinforcement learning** framework for decentralized traffic signal control with shared policy/value networks and explicit constraints on queue growth, unsafe phase transitions, and intersection fairness.
- Trained agents under **non-stationary demand** with **curriculum learning** and **domain randomization**, tracking travel time, queue length, and constraint violations; compared **PPO, MAPPO**, and value decomposition across seeds.

Sensitivity-Aware LLM Quantization and KV-Cache Compression | PyTorch, CUDA, Transformers **Jun 2026**
- Developing post-training **mixed-precision (4/8/16-bit) quantization** that ranks attention heads and layers by activation and perturbation sensitivity, assigning precision to compress model weights and long-context **KV-cache** memory.
- Benchmarking perplexity, downstream accuracy, tokens/second, TTFT, and peak VRAM against FP16 and uniform INT8 baselines; generating **per-layer sensitivity maps** and calibration-coverage diagnostics that explain which layers need higher numerical fidelity.

CTR Prediction Ablation Study (DCN V2) | PySpark, TensorFlow, LightGBM | GitHub **Sep 2026**
- Compared **DCN V2**, LightGBM, and a deep-only ablation on **3M Criteo impressions** with chronological splits, a leakage audit, 3 seeds, **paired bootstrap CIs**, and calibration; LightGBM led DCN V2 (0.688 vs. 0.686 AUC); cross layers added 0.0009 AUC.

Zero-Shot Vision-Language Retrieval | SigLIP, Qdrant, FastAPI | GitHub **Sep 2026**
- Evaluated **zero-shot SigLIP** embeddings for text-to-image retrieval on **COCO val2017** (5,000 images, 25,014 caption queries): **47.2%** / **72.1%** / **80.6%** Recall@1/5/10 and 0.58 MRR, with 6.7 ms p50 vector search.

INDUSTRY EXPERIENCE

Bristol Myers Squibb **Apr 2024 - Jul 2025**
Decision Analyst I - Data Science and AI Solutions *Hyderabad, India*
- Built **Python** scoring pipelines across **5+ sources** (+**15%** forecast accuracy) and **A/B experiments** supporting **$2M+** impact.

ZS Associates **Jul 2023 - Apr 2024**
Decision Analytics Associate - Multi-Channel Marketing *Pune, India*
- Built **SQL, R, and Python** feature pipelines and predictive models, cutting refresh time by **30%** and manual effort by **20%**.

TECHNICAL SKILLS & AWARDS

Languages & Frameworks: Python, C++, SQL, R, PyTorch, TensorFlow, Hugging Face, LoRA/QLoRA, Ray RLlib, vLLM, CUDA
Models & Methods: Transformers, LLMs, VLMs, CNN/LSTM, RL (PPO, MAPPO), curriculum learning, quantization
Evaluation: zero-/few-shot and red-team evaluation, ablation design, calibration, bootstrap CIs, A/B testing, SHAP
Compute & Tools: HiPerGator HPC (multi-GPU), Docker, MLflow, PySpark, FastAPI, Qdrant, pgvector, Git, Linux
Awards: Winner, Best Use of Orkes, UC Berkeley AI Hackathon 2026 (BrailleAI); MLH Winner, Gator Quant Hacks